

Robust Causal Inference Under Informative Missingness

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Introduction

- Missing data is common in clinical and observational studies and can bias causal effect estimation if handled improperly¹
- Standard approaches rely on strong assumptions and their performance under missing data conditions remains inadequately quantified
- Existing literature often evaluates methods in isolation, with limited simulation evidence comparing causal estimators under realistic clinical data structures²

Objective

- To evaluate causal methods under different missing data mechanisms (MCAR, MAR, MNAR) and their impact on causal effect estimation in survival analysis settings

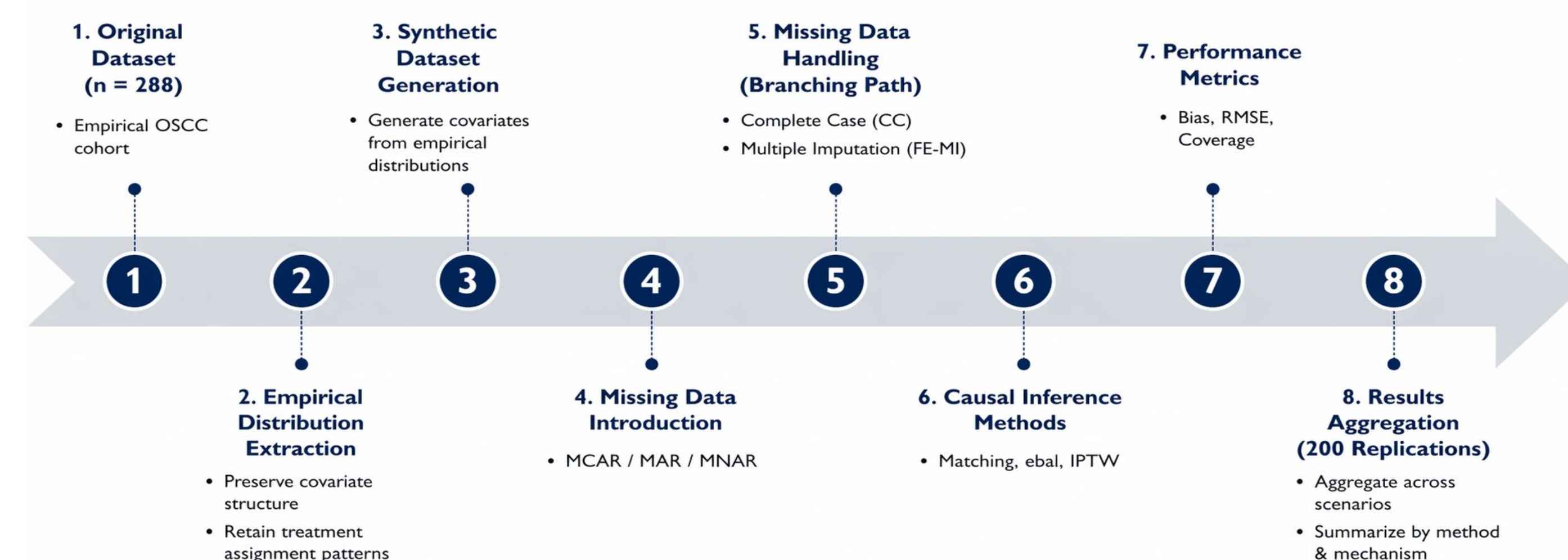
Simulation Study

- Empirical distribution based on oral squamous cell carcinoma (OSCC) cohort of n=288³
- Treatment of interest is the engraftment of tumours which is confounded with many covariates
- Simulated datasets generated to preserve realistic covariate and treatment structure
- Survival outcomes include disease-free survival (DFS) simulated using a Weibull model
- Missingness is introduced in the two strongest confounders to assess the strength of the causal methods
- 200 replications performed per condition

Simulation Design	Condition
Sample Size	500, 1000, 2000
True HR	1.0, 1.5, 2.0
Missingness Proportion	0%, 25%, 50%
Causal Method	IPTW, ebal, matching
Missing Data Strategy	Complete case, Multiple Imputation
Missing Data Mechanism	MCAR, MAR, MNAR

Methods

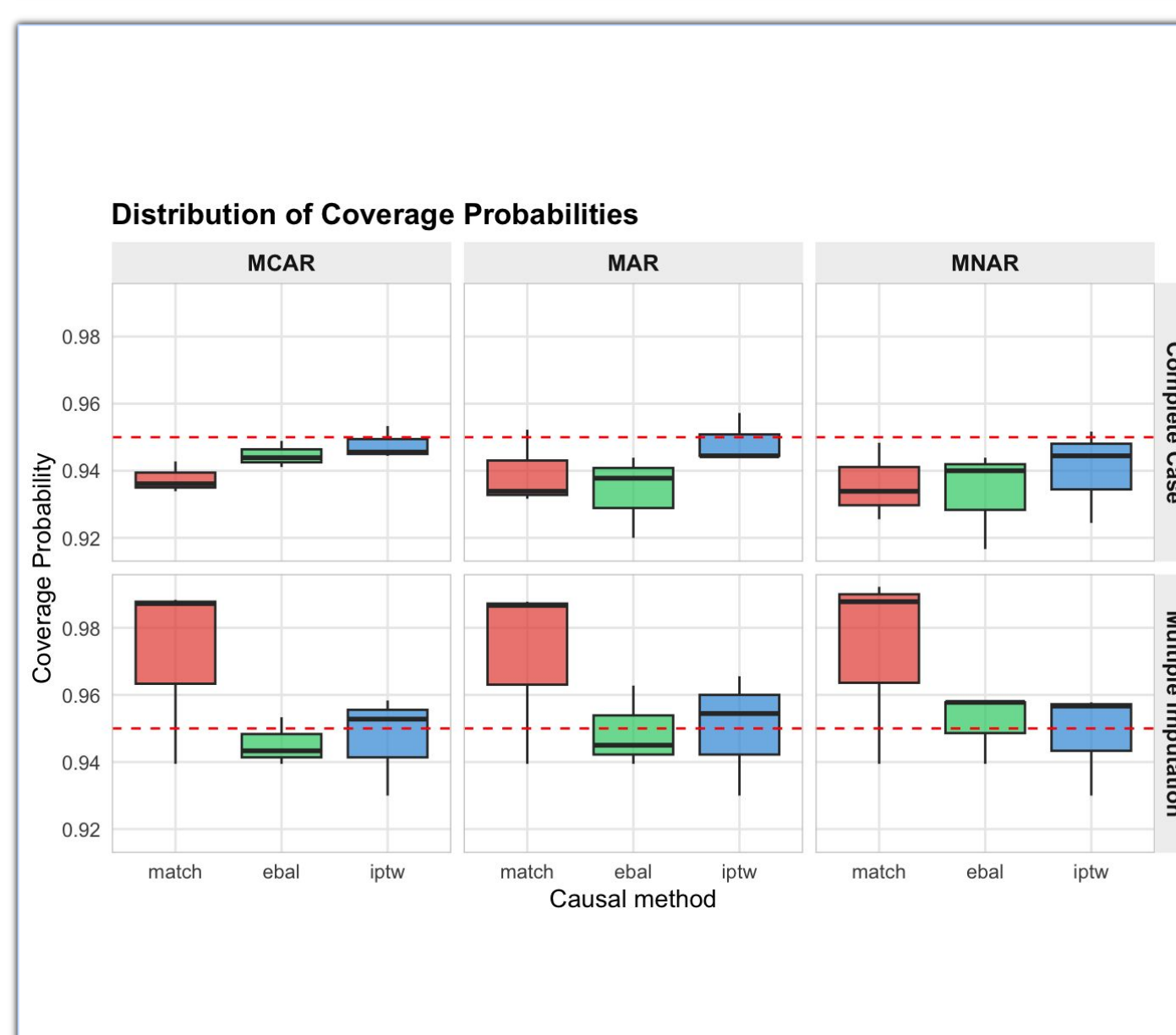
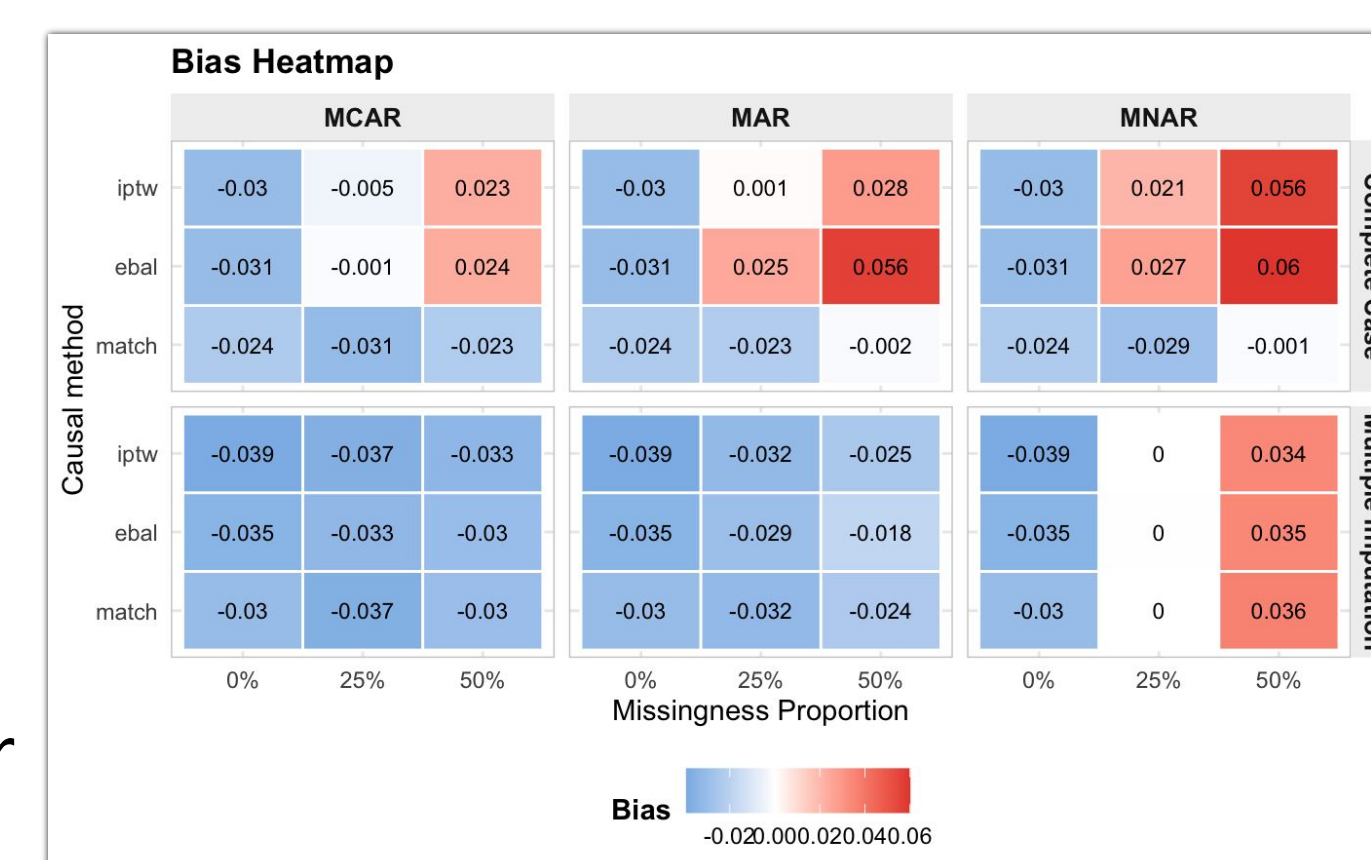
- The causal estimand is the Average Treatment Effect on the Treated (ATT), targeting the treatment effect among engrafted patients to reflect settings where treatment assignment is non-random and inference is focused on treated patients
- Causal inference methods include inverse probability treatment weighting (IPTW), entropy balancing (ebal) and propensity score matching (full matching)
- Performance is evaluated through bias (log hazard ratio), root mean square error (RMSE) and 95% confidence interval coverage of the hazard ratio



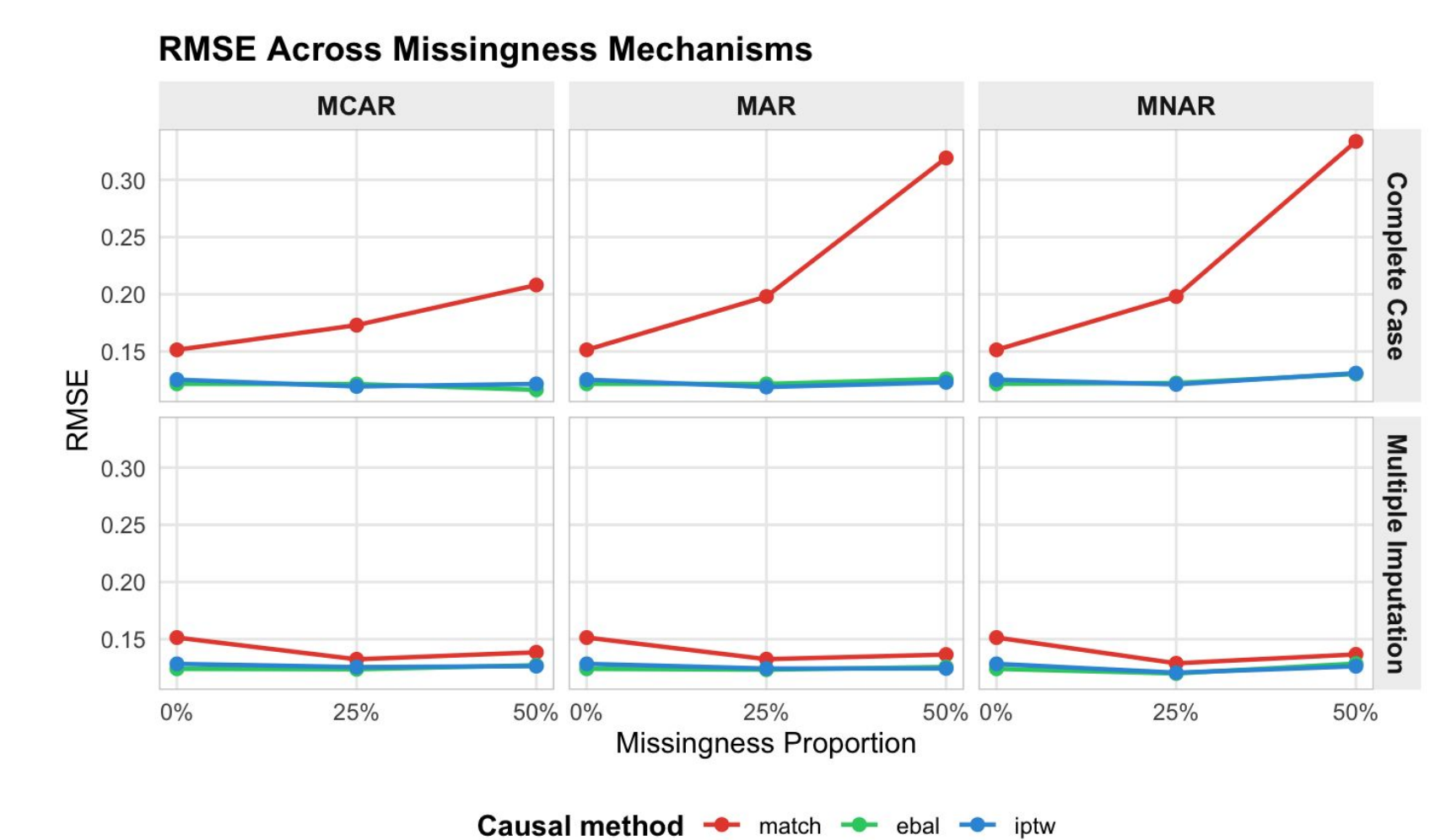
Preliminary Results

- Across all simulations, missing data handling strategy dominated causal method selection, with substantially larger performance differences between complete case (CC) and multiple imputation (MI) than between matching, IPTW, and entropy balancing
- Performance deterioration was most pronounced under MAR and MNAR mechanisms

- Under complete case analysis, bias increases in magnitude with missingness, with IPTW and entropy balancing showing upward bias (overestimation) at higher missingness levels
- Multiple imputation substantially reduced bias across all methods, yielding near-zero bias even at higher missingness levels.



- Complete case analysis shows undercoverage relative to the nominal 95% level with greater deterioration under MNAR
- Coverage under MI was at or above nominal levels, though:
 - Propensity score matching showed inflated coverage suggesting conservative variance estimation
 - IPTW and entropy balancing achieved coverage closer to nominal



- Under complete case analysis, RMSE increases with missingness, particularly for matching, which shows substantial variability under MAR and MNAR
- Multiple imputation substantially reduces RMSE across all methods, with minimal differences between causal approaches

Conclusions

- Missing data handling is the primary driver of performance, with larger effects than the choice of causal method
- Complete case analysis leads to bias, undercoverage, and increased variability, particularly under MAR and MNAR mechanisms
- Multiple imputation provides robust inference, eliminating bias and restoring coverage across all methods
- Differences between causal methods are minimal after imputation, indicating that performance is largely driven by missing data
- Propensity score matching exhibits overcoverage under MI, suggesting conservative variance estimation

Next Steps

- Extend to simulation to include other outcomes such as overall survival (OS) which incorporate correlated DFS and OS outcomes to better reflect clinical settings
- Investigate matching overcoverage and assess impact of matching design choices
- Examine variance estimation and effective sample size
- Expand simulation scenarios to include larger sample sizes to investigate their impact
- Evaluate performance across larger sample sizes
- Explore more complex MAR and MNAR mechanisms and drivers

References:

- Mayer et al. 2023. Biometrical Journal, 65(5). <https://doi.org/10.1002/bimj.202100294>
- Rippin, G., et al. 2024. Drug Safety, 47(12). <https://doi.org/10.1007/s40264-024-01345-3>
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